



**Sultan Qaboos University**  
**College of Engineering**  
 Office of Assistant Dean for  
 Industrial Training and Community Services

Weekly Report

Week #: 8

Date:

|                                                                                                                                                                                                           |                                                                                                                                                                         |                                    |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|
| Student                                                                                                                                                                                                   | Full Name: <u>Malek Al Shukily</u>                                                                                                                                      | ID: <u>141776</u>                  |
|                                                                                                                                                                                                           | Department: <input type="checkbox"/> CAE <input type="checkbox"/> ECE <input type="checkbox"/> MIE <input type="checkbox"/> PCE <input checked="" type="checkbox"/> MCE | Mobile: <u>72471416</u>            |
| Supervisor                                                                                                                                                                                                | Full Name: <u>Nasser Zahir Al-Aufi</u>                                                                                                                                  | Email: <u>nzs960fi@hotmail.com</u> |
|                                                                                                                                                                                                           | Training Company & Department:<br><u>MODES / Drilling Sect</u>                                                                                                          | Tel./Mobile:<br><u>24339162</u>    |
| Tasks Completed (Briefly Describe any tasks that you completed during this week):                                                                                                                         |                                                                                                                                                                         |                                    |
| <ul style="list-style-type: none"> <li>• We learned the types of submersible and surface pumps.</li> <li>• We learned the stages of well drilling.</li> <li>• We learned how to inspect pumps.</li> </ul> |                                                                                                                                                                         |                                    |
| Tasks in Progress (if any):                                                                                                                                                                               |                                                                                                                                                                         |                                    |
| <ul style="list-style-type: none"> <li>• —</li> <li>• —</li> <li>• —</li> </ul>                                                                                                                           |                                                                                                                                                                         |                                    |
| Plan for next week (if different from this week):                                                                                                                                                         |                                                                                                                                                                         |                                    |
| <ul style="list-style-type: none"> <li>• We finish the training for two months.</li> <li>• —</li> <li>• —</li> </ul>                                                                                      |                                                                                                                                                                         |                                    |
| Problems/Challenges/Recommendations (if any):                                                                                                                                                             |                                                                                                                                                                         |                                    |
| <u>No Major Problems we Phase</u>                                                                                                                                                                         |                                                                                                                                                                         |                                    |
| Supervisor Comments (your feedback is very important for our continuing improvement):                                                                                                                     |                                                                                                                                                                         |                                    |
| <u>No Comment</u>                                                                                                                                                                                         |                                                                                                                                                                         |                                    |

**Student:**

I have read and understood the whole content of the Students' Training Manual.

Signature: Malek Al Shukily**Supervisor:**

I will do my best to guide the student towards achieving all required deliverables.

Signature: Nasser Zahir Al-Aufi

**Note:** A copy of this report should be faxed to 24413416 or scanned and emailed to your SQU supervisor on a weekly basis. Originals should be placed in the Appendix of the Final Report.



# WEEKLY OPERATIONAL REPORT

Water Well Drilling & Pump Systems Department

WEEK 33 (AUG 07 – AUG 13, 2026)

Report ID: WWR-2026-W33

Prepared By: Operations Manager

Department: Hydrogeology & Pumping Solutions

Status: Approved

## EXECUTIVE SUMMARY

During Week 33, operational efficiency across drilling and pumping operations reached **94.2%**. The department successfully completed **3 water well drilling projects** to target depths, installed **5 high-efficiency submersible pump systems**, and carried out scheduled preventive maintenance on **8 surface booster pumps**. Zero Lost Time Injuries (LTI) were recorded, achieving 180 consecutive incident-free operational days.

3

WELLS DRILLED

↑ 100% Target Met

620m

TOTAL METERAGE

+45m vs Target

5

SUBMERSIBLE  
PUMPS

4 New / 1 Overhaul

8

SURFACE PUMPS

100% Serviced

0

LTI / SAFETY  
INCIDENTS

180 Days Safe

## 1. WATER WELL DRILLING OPERATIONS

| RIG ID       | LOCATION / SITE           | TARGET DEPTH | DRILLED DEPTH | EST. YIELD (M³/H) | STATUS             |
|--------------|---------------------------|--------------|---------------|-------------------|--------------------|
| Rig-01 (Mud) | North Sector - Farm A2    | 220 m        | 220 m         | 45 m³/h           | Completed & Cased  |
| Rig-02 (DTH) | Central District - Site B | 180 m        | 180 m         | 60 m³/h           | Completed & Tested |
| Rig-03 (DTH) | East Industrial Zone      | 250 m        | 220 m         | Pending Test      | In Progress (88%)  |

## 2. PUMP INSTALLATION & MAINTENANCE OPERATIONS

### A. Submersible Pumps (Deep Well & Aquifer Extraction)

| SITE / WELL ID   | PUMP SPEC / BRAND             | POWER / HEAD      | ACTIVITY PERFORMED               | STATUS                     |
|------------------|-------------------------------|-------------------|----------------------------------|----------------------------|
| Well-A2 (North)  | Grundfos 6" Multi-Stage       | 30 HP / 180m Head | New Installation & Riser Piping  | Commissioned               |
| Well-B (Central) | Caprari High-Flow Submersible | 45 HP / 140m Head | New Installation & VFD Setup     | Commissioned               |
| Well-W8 (South)  | KSB Deep Well Pump            | 25 HP / 120m Head | Motor Rewinding & Cable Splicing | Re-installed & Operational |
| Well-E3 (East)   | Lowara Stainless Submersible  | 20 HP / 110m Head | Drop Pipe Replacement & Testing  | Final Testing              |

B. Surface & Booster Pumps (Distribution & Irrigation)

| EQUIPMENT ID | PUMP TYPE                          | MAINTENANCE SCOPE                             | PERFORMANCE CHECK       | STATUS        |
|--------------|------------------------------------|-----------------------------------------------|-------------------------|---------------|
| SP-BS-01     | Horizontal End-Suction Centrifugal | Mechanical seal replacement & alignment       | Flow & Vibration Normal | Operational   |
| SP-MS-04     | Vertical Multistage Booster        | Bearing repacking & impeller inspection       | Pressure Test Passed    | Operational   |
| SP-IRR-02    | Diesel-Driven Surface Pump         | Oil change, filter replacement & priming val. | RPM & Pressure Normal   | Standby Ready |

Health, Safety & Environment (HSE)

- **Safe Work Hours:** 1,120 man-hours worked this week without any incident or near-miss.
- **Toolbox Talks Held:** Rig positioning safety, electrical safety during high-voltage pump testing, and heavy pipe lifting hazards.
- **Environmental Management:** 100% containment of drilling mud fluids at Rig-01 site with zero ground spill.

Challenges & Mitigation Actions

- **Hard Stratum Delay (Rig-03):** Encountered high-hardness basalt layer at 180m depth. *Mitigation:* switched to heavy-duty DTH button bit.
- **Voltage Fluctuation (Site B):** Power instability affecting pump VFD commissioning. *Mitigation:* Installed dedicated AVR system.

3. PLANNED SCHEDULE FOR NEXT WEEK (WEEK 34)

| TARGET AREA            | PLANNED TASK / SCOPE                                           | ASSIGNED RIG / CREW   | TARGET DATE  |
|------------------------|----------------------------------------------------------------|-----------------------|--------------|
| East Industrial Zone   | Complete Rig-03 drilling to 250m & execute yield pump test     | Rig-03 Drilling Team  | Aug 16, 2026 |
| South Field - Sector C | Mobilization of Rig-01 for new 200m agricultural well          | Rig-01 / Rigging Crew | Aug 17, 2026 |
| Main Distribution Hub  | Installation of 2 x 55HP High-Pressure Surface Booster Pumps   | Pump Tech Team A      | Aug 18, 2026 |
| Farm A2                | Final commissioning & handover of submersible system to client | Commissioning Eng.    | Aug 19, 2026 |

|                                                                                                                             |                                                                                                                                                       |
|-----------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>Prepared By:</b></p> <p>Eng. Robert Vance<br/>Senior Operations Engineer</p> <p>-----</p> <p>Signature &amp; Date</p> | <p><b>Reviewed &amp; Approved By:</b></p> <p>Eng. Tariq Al-Mansoori<br/>Department Head - Water Projects</p> <p>-----</p> <p>Signature &amp; Date</p> |
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